

Sales & Operations Planning

A workshop on:

- ***what it is***
- ***how it can help***
- ***how to make it work***

by

Tom Wallace

for

OMTEC – Chicago – June 16, 2010

Objectives of this Session

Enable each attendee to:

- Understand where Executive S&OP fits, what it does, why it's important
- Understand and explain the logic and mechanics of Executive S&OP
- Explain the elements of the monthly process
- Be exposed to a highly effective implementation process.
- Other?

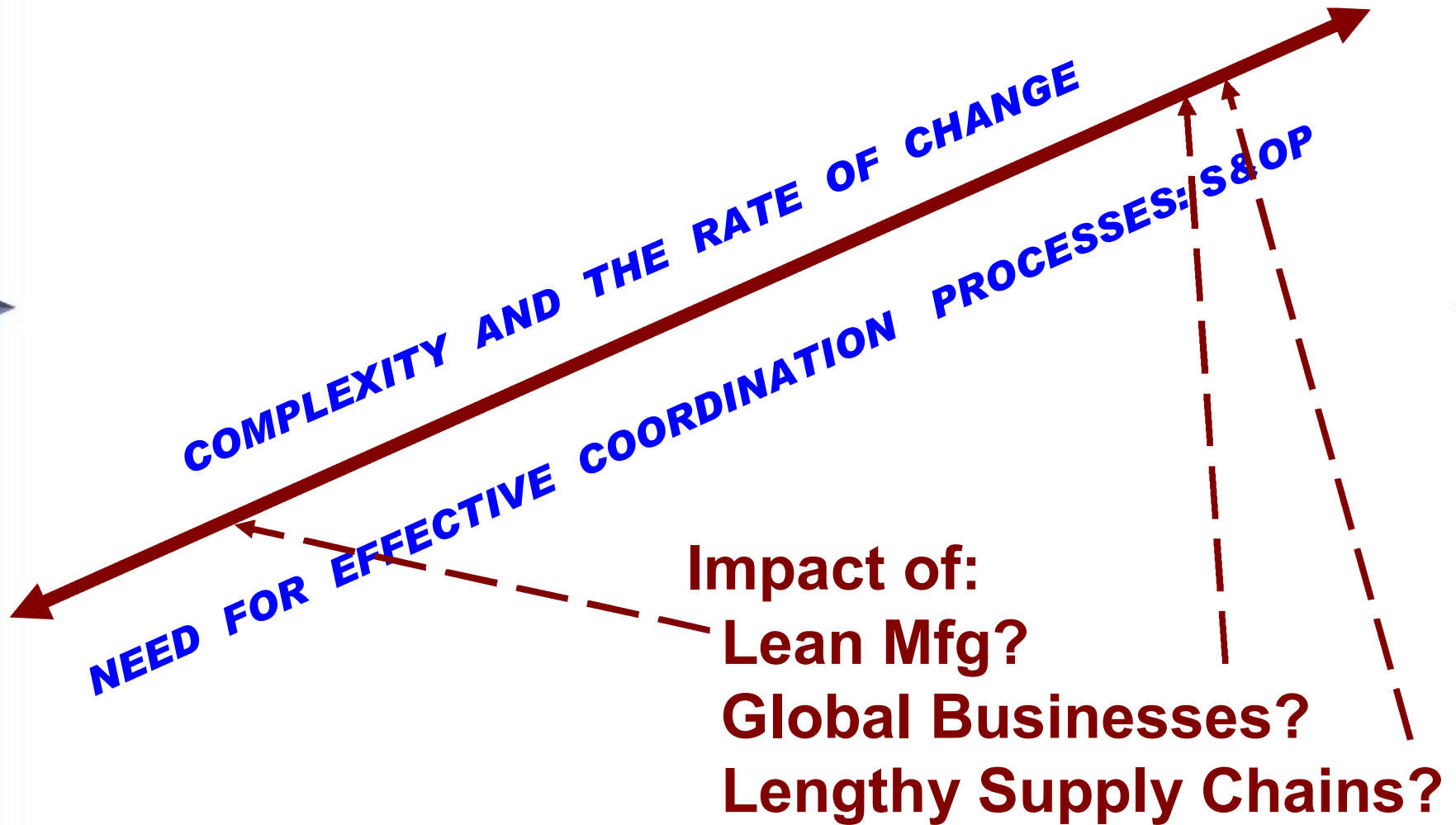
Where is Your Company Relative to Executive SOP?

- 1 – Successfully implemented
- 2 – Implemented but not working well
- 3 – Implementing now
- 4 – Not yet implementing

Outline XXXXX

- Sales & Operations Planning
Role, Scope, Benefits
- Display of S&OP Information
- The Executive S&OP Process
 - Demand Planning (Forecasting)
 - Supply (Capacity) Planning
 - Pre-meeting
 - Exec Meeting
- Global Executive S&OP
- Risk Management
- Implementation
- Problem Solving Session
- The Future of Sales & Operations Planning

Complexity, Change and Coordination



EXPERT INPUT

ADAM SZCZEPANSKI

Vice President, Corporate Strategy and Operations
Forum Oil Technologies

Annual Sales: ~ \$400 million



Main Products: Capitol Equipment and Tooling for drilling contractors

Alan was formerly CFO, V&M Star (another oil field equipment company)

EXPERT INPUT

CONOR McDONALD

Supply Chain Manager
Microsoft Corporation

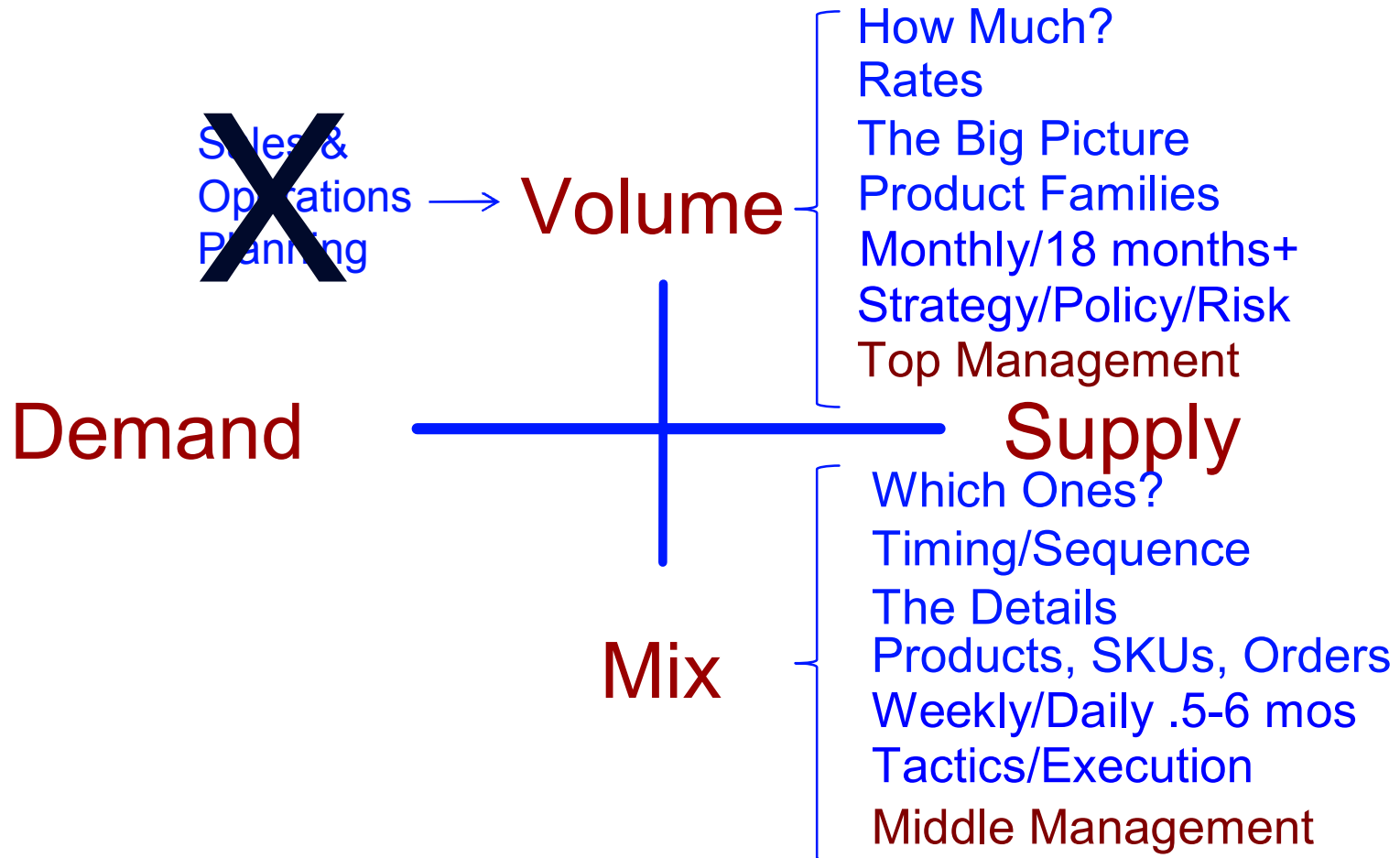
Annual Sales: ~ \$50 billion

Main Products: Software, Games,
and Computer Accessories

Conor is based in Ireland



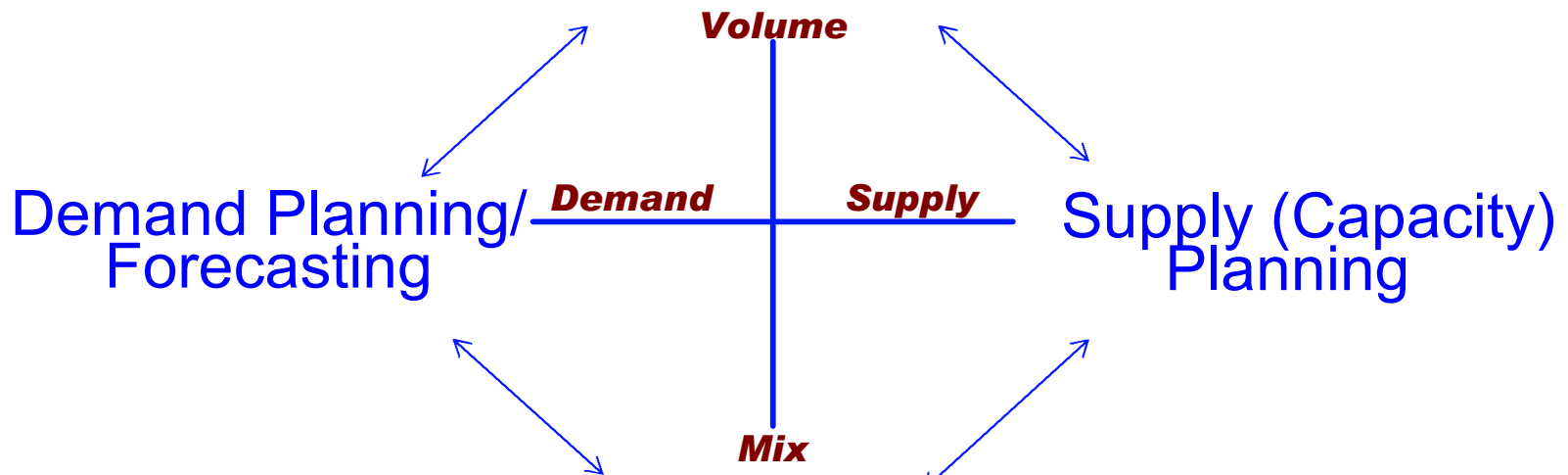
The Four Fundamentals



Sales & Operations Planning

Executive S&OP

Volume Planning for Demand and Supply
Medium to Long-Term Financial Planning
Support for Earnings Calls, Risk Mgmt, Strategy



Operational S&OP

Master Scheduling
Plant, Supplier and Distribution Scheduling
(Kanban, APS, etc.)

Slide 9

b6

I think the key thing to hit is "what is Executive S&OP"? Should we have some bullet points somewhere that give some basic description? And can we link this back to some key points:?

- 1) This is about Executive" Decision Making"
- 2) This is about" Business Performance" (not just demand/supply balancing)
- 3) This is not "detailed execution review"

bodell, 8/16/2006

Executive S&OP . . .

- Is an executive decision-making process
- Balances demand and supply
- Deals with volume in both units and \$\$\$
- Links operational plans to financial plans: one set of numbers
- facilitates risk management
- Is the forum for setting relevant strategy and policy

Executive S&OP . . .

. . . is essential for the elements within
Operational S&OP –

Master Scheduling

Distribution Requirements Planning

Plant Scheduling – APS, Kanban, etc.

Supplier Scheduling

– to function well and contribute to their
full potential.

From 5,000 Feet

Sales/Marketing
(Units/\$\$\$ by Family)

Finance
(Dollars)

Executive S&OP

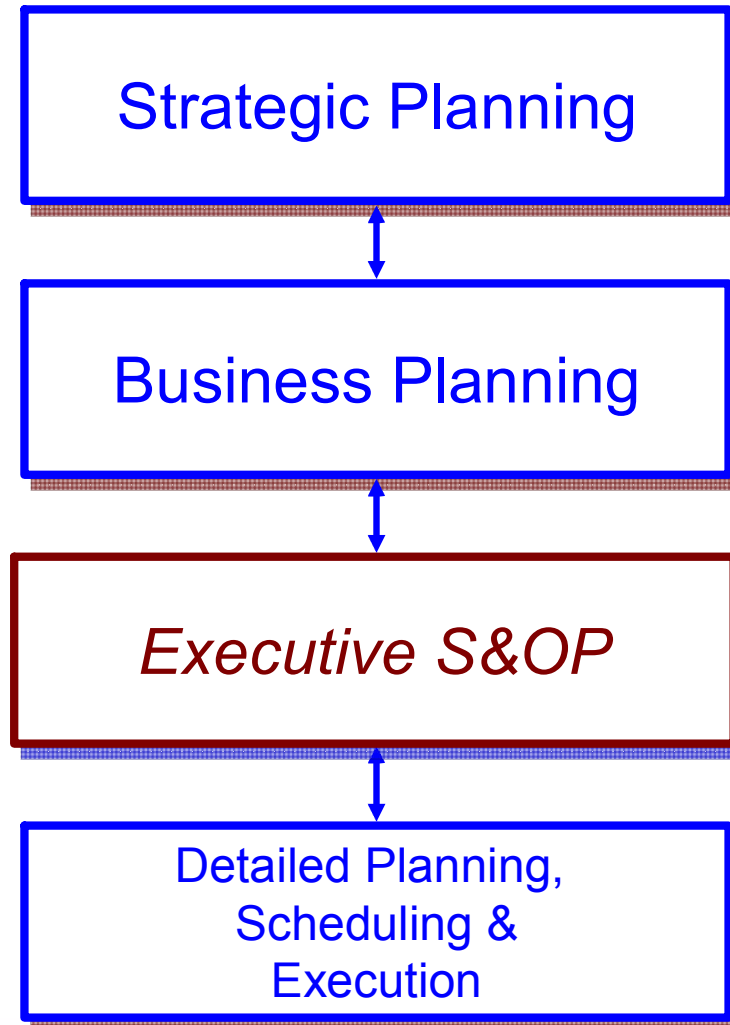
A process to reconcile, agree upon, and communicate the company game plan --

*Centripetal
Shared View
Teamwork*

Operations
(Units/hours/Material)

Product Development
(New Product Issues)

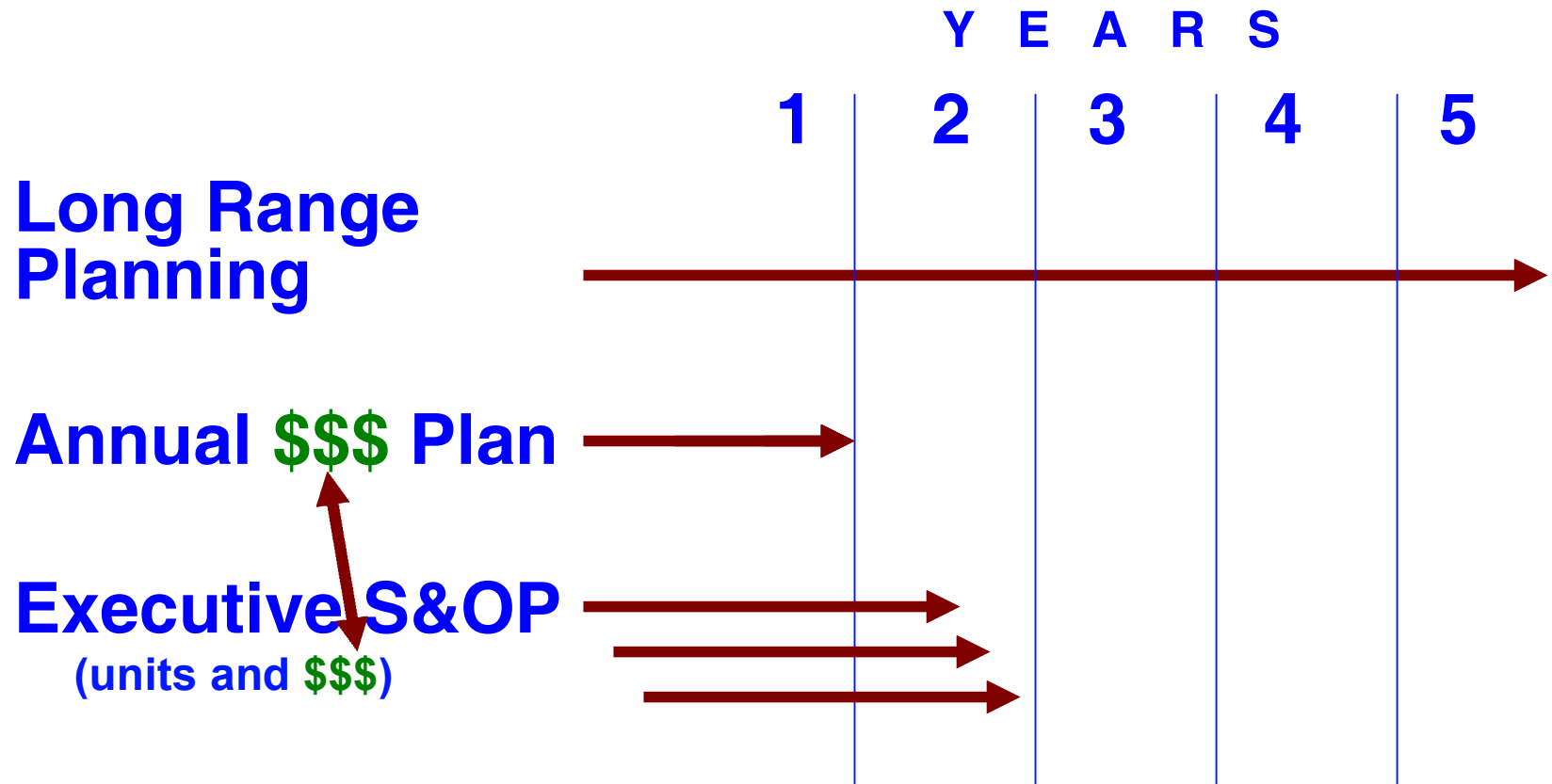
Where Executive S&OP Fits



Business Plan Characteristics

- Dollars
- 3 to 5 Years
- Annual Budget
- Entire Business
- Ownership

Levels of Business Planning



Rolling monthly check on the Business Plan

The Role of Top Management

The Leader of the Business Unit (General Manager, President, COO) Needs to be Hands-On with Executive S&OP:

1. Stewardship

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graph TD; A[Strategic Planning] <--> B[Business Planning]; B <--> C["Disconnect!!!"]; C <--> D["Detailed Planning, Scheduling & Execution"]
```

Strategic Planning

Business Planning

Disconnect!!!

Detailed Planning,
Scheduling &
Execution

The Role of Top Management

The Leader of the Business Unit (General Manager, President, COO) Needs to be Hands-On with Executive S&OP:

1. Stewardship
2. Leadership
 - Break ties

The Role of The President (CEO, COO, GM, MD, etc.)

Sales/Marketing
(Units/\$\$\$ by Family)

Finance
(Dollars)



Operations
(Units/hours/Material)

Product Development
(New Product Issues)

The Role of Top Management

The Leader of the Business Unit (General Manager, President, COO) Needs to be Hands-On with Executive S&OP:

1. Stewardship
2. Leadership
 - Break ties
 - Set high standards
 - Motivate

Monthly Time Commitment: **1.5 hours**

Hard Benefits

- | | |
|-------------------------------|------|
| • Customer Service | UP |
| • Inventory | DOWN |
| • Obsolescence | DOWN |
| • Freight Costs | DOWN |
| • Order Lead Times | DOWN |
| • Supplier Lead Times | DOWN |
| • Time to Launch New Products | DOWN |
| • Plant Productivity | UP |

EXPERT INPUT

CATHY BUDD

Director, Demand Driven Business Operations
The Dow Chemical Company

Annual Sales: ~ \$54 billion

Main Products: Chemicals

Cathy serves as a senior consultant to operating businesses within Dow



EXPERT INPUT

DEAN SMETANA

Director of Materials, North America
Office Products of Newell Rubbermaid

Annual Sales: ~ \$1 billion

Main Products: Pens & Pencils, Desk Accessories



Soft Benefits

- Enhanced Teamwork
- Improved Communications – Institutionalized
- Better Decisions with Less Effort and Time
- Better \$\$\$ Plans with Less Effort and Time
- Greater Accountability
- Greater Control
- Window into the Future

Top Management's Handle on the Business

EXPERT INPUT

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Director of Materials, North America
Office Products of Newell Rubbermaid

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Main Products: Pens & Pencils, Desk Accessories



Bad Day at Acme Widget

Product Manager, reviewing forecast for 1st time in 3 mos.

	<u>Feb</u>	<u>Mar</u>	<u>Apr</u>		<u>May</u>	<u>Jun</u>	<u>Jul</u>	<u>Aug</u>	<u>Sep</u>
Forecast (000)	100	100	100		100	120	120	120	120
Actual Sales	90	95	85						
Difference	-10	-5	-15						
Cum Difference		-15	-30						

Product Manager revises the forecast downward.

	<u>Feb</u>	<u>Mar</u>	<u>Apr</u>		<u>May</u>	<u>Jun</u>	<u>Jul</u>	<u>Aug</u>	<u>Sep</u>
Forecast (000)	100	100	100	Old Forecast	100	120	120	120	120
				New Forecast	90	90	90	90	90
Actual Sales	90	95	85						
Difference	-10	-5	-15						
Cum Difference	-10	-15	-30						

Production Plan

Current Production Plan.

	<u>Feb</u>	<u>Mar</u>	<u>Apr</u>		<u>May</u>	<u>Jun</u>	<u>Jul</u>	<u>Aug</u>	<u>Sep</u>
Plan Production (000)	100	100	100		110	120	120	120	120
Actual Production	98	100	101						
Difference	-2	0	+1						
Cum Difference	-2	-2	-1						

Plant Manager gets bad news & adjusts Production Plan.

	<u>Feb</u>	<u>Mar</u>	<u>Apr</u>		<u>May</u>	<u>Jun</u>	<u>Jul</u>	<u>Aug</u>	<u>Sep</u>
Plan Production (000)	100	100	100	Old Plan	110	120	120	120	120
				New Plan	110	100	100	90	90
Actual Production	98	100	101						
Difference	-2	0	+1						
Cum Difference	-2	-2	-1						

Inventory Review

CFO observes the high inventories.

	<u>Feb</u>	<u>Mar</u>	<u>Apr</u>	<u>May</u>	<u>Jun</u>	<u>Jul</u>	<u>Aug</u>	<u>Sep</u>
Plan Inventory (1 Mo)	100	100	100	120	120	120	120	120
Actual Inv.	103	111	116					
Difference	+11	+16	+32					

Average Standard Cost ~ \$100 per unit
April Ending Inventory ~ \$13,200,000

Inventory Review

CFO gets new Forecast and Production Plan.

	<u>Feb</u>	<u>Mar</u>	<u>Apr</u>		<u>May</u> 90	<u>Jun</u> 90	<u>Jul</u> 90	<u>Aug</u> 90	<u>Sep</u> 90
Forecast (000)				New Forecast					
Actual Sales	90	95	85						
Difference	-10	-5	-15						
Cum Difference	-10	-15	-30						
Plan Production				New Plan	110	100	100	90	90
Actual Production	98	100	101						
Difference	-2	0	+1						
Cum Difference	-2	-2	-1						

Current Inventory 132

New Inv. Plan 152 162 172 - - - ➤

Sally Smith's Spreadsheet

		FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP
FORECAST	100	100	100	100	120	120	120	120	
ACTUAL SALES	90	95	85						
DIFFERENCE	-10	-5	-15						
CUM DIFFERENCE	-15	-30							
PLANNED PRODUCTION	100	100	110	120	120	120	120		
ACTUAL PRODUCTION	98	100	101						
DIFFERENCE	-2	0	+1						
CUM DIFFERENCE		-2	-1						
PLANNED INV (1 MONTH SUPPLY)	100	100	100	???					
ACTUAL INVENTORY (JAN=103)	111	116	132						
DIFFERENCE	+11	+16	+32						

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PLANNED PRODUCTION	100	100	100	110	120	120	120	120	
ACTUAL PRODUCTION	98	100	101						
DIFFERENCE	-2	0	+1						
CUM DIFFERENCE		-2	-1						

PLANNED INV (1 MONTH SUPPLY)	100	100	100	142	142	142	142	142	
ACTUAL INVENTORY (JAN=103)	111	116	132						
DIFFERENCE	+11	+16	+32						

Is this plan okay?

Sally Smith's Spreadsheet

		FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP
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ACTUAL SALES	90	95	85						
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PLANNED PRODUCTION	100	100	100	110	110	120	120		
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PLANNED INVENTORY (1-MO)	100	100	100	142					
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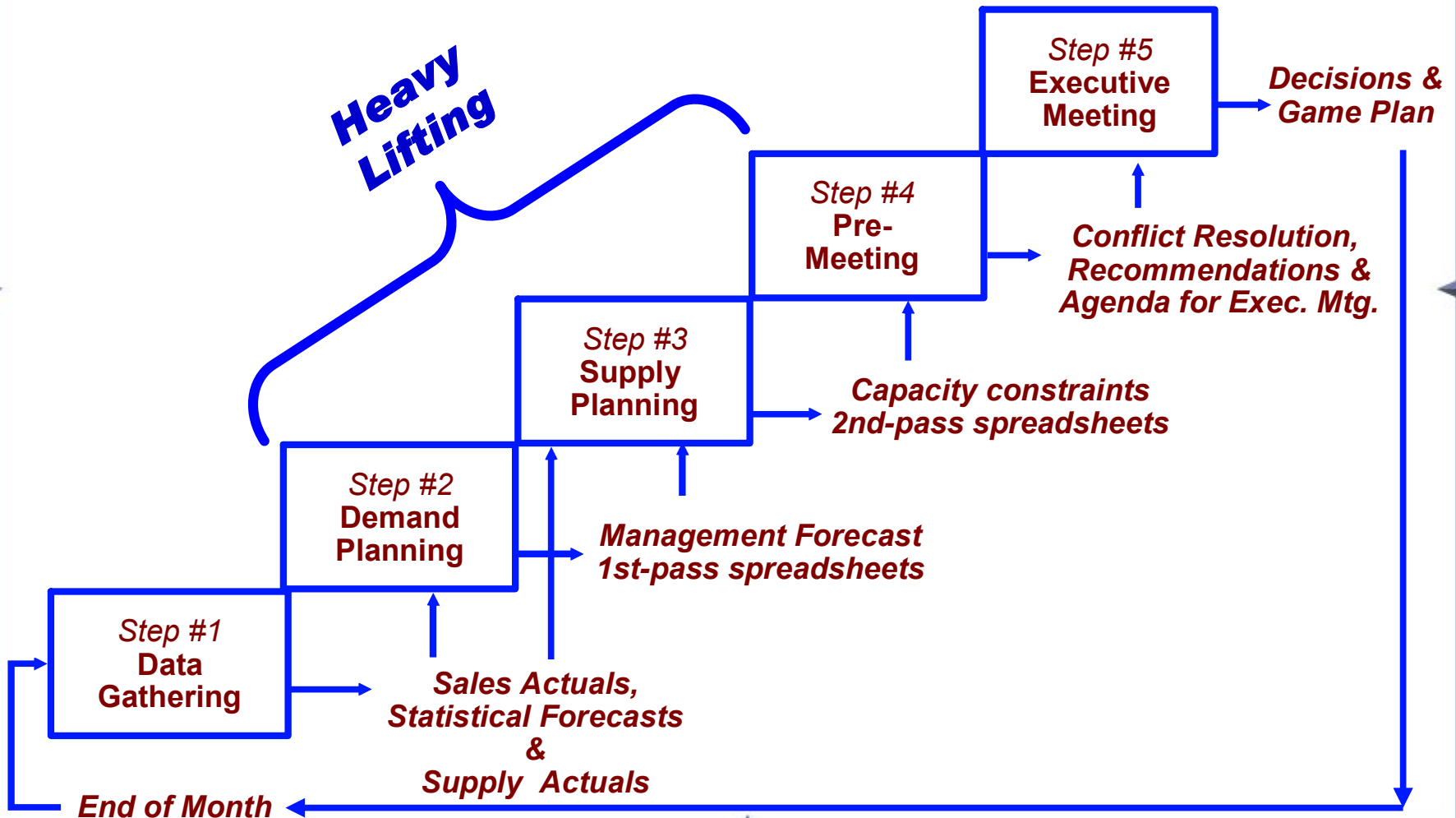
PLANNED PRODUCTION	100	100	100	110	110	120	120		
ACTUAL PRODUCTION	98	100	101						
DIFFERENCE	-2	0	+1						
CUM DIFFERENCE		-2	-1						

PLANNED INVENTORY (1-MO)	100	100	100	142	132	122	122	122	
ACTUAL INVENTORY (JAN=103)	111	116	132						
DIFFERENCE	+11	+16	+32						

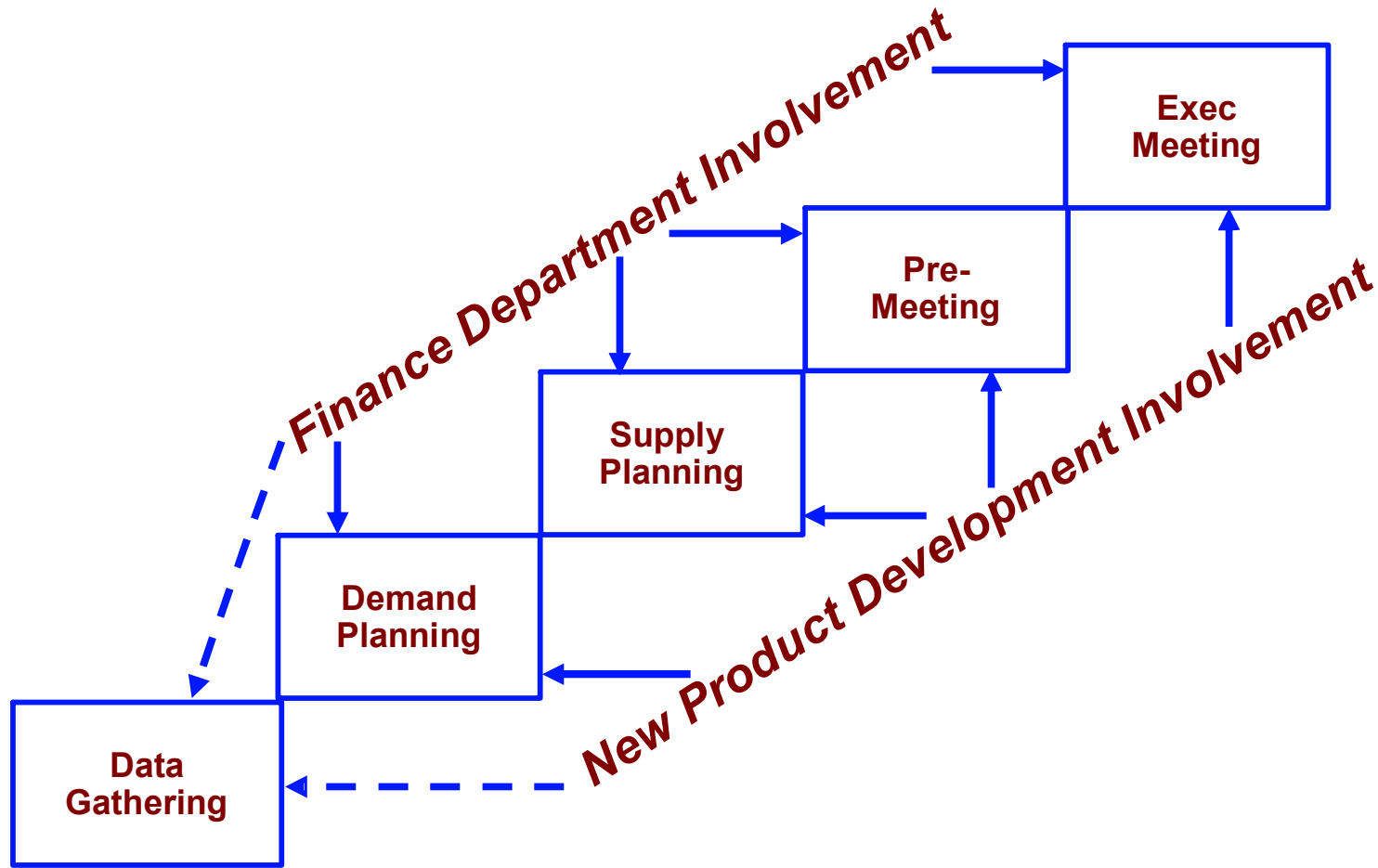
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- Sales & Operations Planning
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The Executive S&OP Process



Role of Finance and New Product Development



EXPERT INPUT

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Vice President, Corporate Strategy and Operations
Forum Oil Technologies

Annual Sales: ~ \$400 million

Main Products: Capitol Equipment and Tooling for drilling contractors

Alan was formerly CFO, V&M Star (another oil field equipment company)



EXPERT INPUT

BRIAN HARLIN

Director of Demand and Fulfillment
Johnsonville Sausage LLC

Annual Sales: ~ \$500 Million

Main Product: Sausage

Brian's Prior Company: a semi-conductor manufacturer



Alternative Scenarios

Situation: Indiana plant is heavily overloaded, can't meet the newly-increased demand plan

Alternatives

- a. Add a shift at Indiana
- b. Offload to plant in Utah
- c. Outsource
- d. Combination

Cost Issues

Supervision,
hiring, training

Freight, cost difference

Total cost to acquire

Most or all of the above

Alternative Scenarios

Alternatives

- a. Add a shift at Indiana
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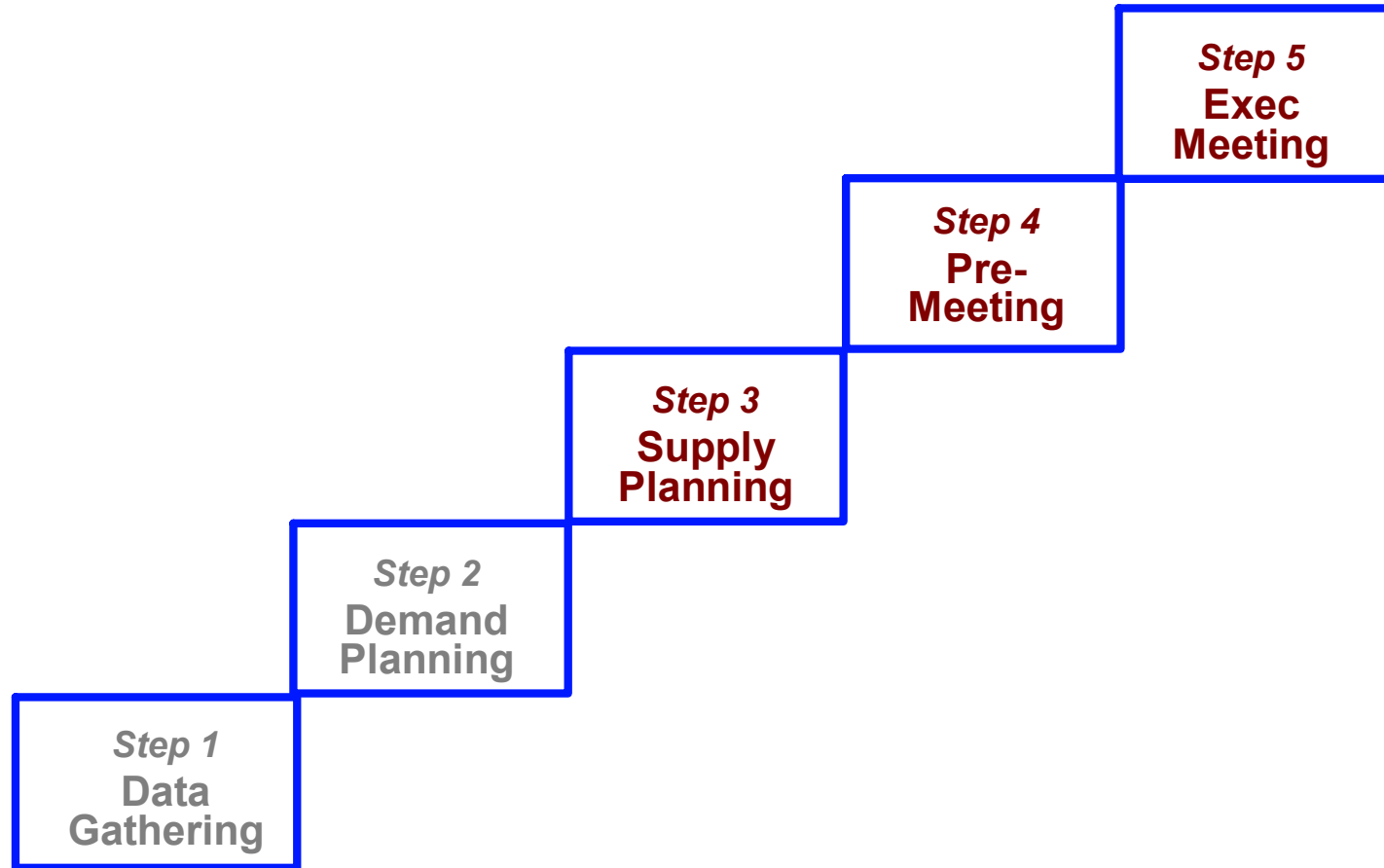
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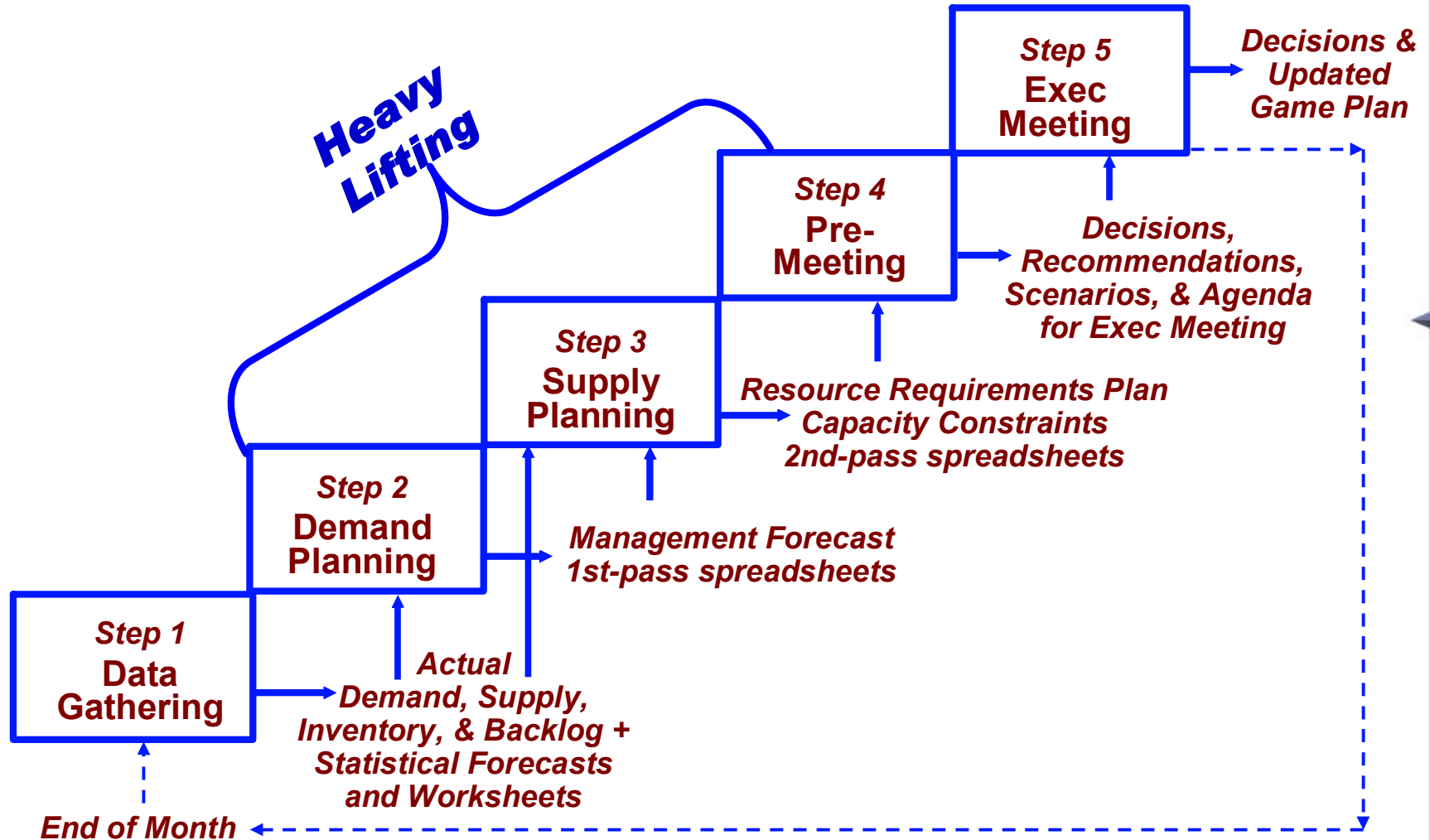
Other implementation issues: time, duration

Other operational issues: quality, lead time,
flexibility etc.

Where Will The Decision Be Made?



The 5-Step Executive S&OP Process



Sample Agenda – Exec Meeting

- Macro \$\$\$ Review – All Families Combined
- Customer Service Performance – Order Fill
- New Product Issues
- Family-by-Family Review and Decisions
- Production/Procurement Rate Changes
- Collective Impact on Business Plan
- Recap of Decisions Made
- Critique of Meeting

Exec Meeting Participants

Essential:

- SBU Leader (Pres./GM)
- VP Sales
- VP Marketing
- VP Operations/Logistics
- VP Product Development
- VP Finance

Probable:

- Exec. S&OP Process Owner
- Pre-Meeting Facilitator
- Demand Manager
- Supply Chain Manager

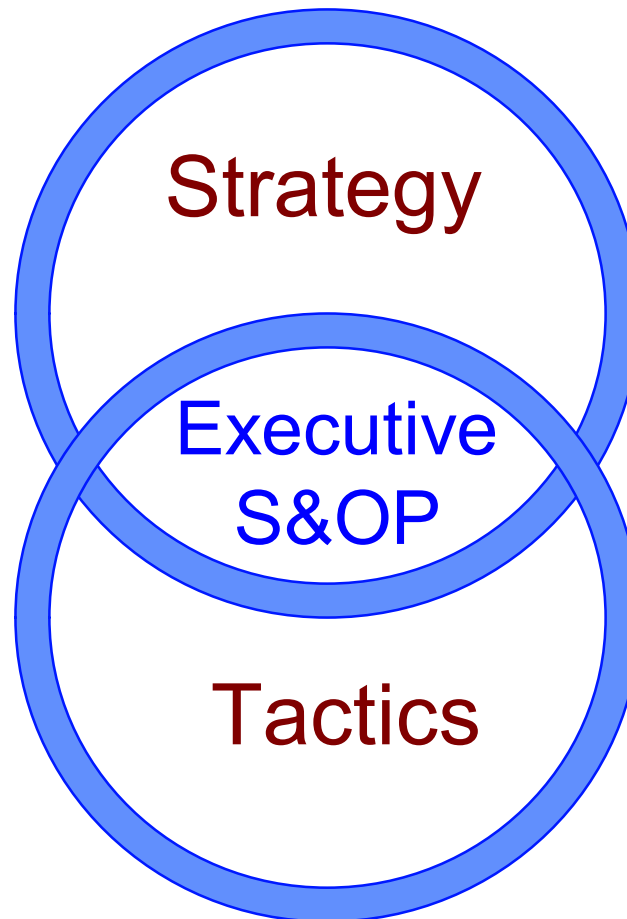
Possible:

- Controller
- Customer Service Manager
- Forecast Analyst
- Human Resource Manager
- Master Scheduler
- Plant Manager
- Product Development Manager
- Product Manager
- Purchasing Manager
- ????

Executive S&OP Timetable

	Working Day of the Month														
<u>Step</u>	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Exec. Meeting															●
Pre-Meeting													●		
Supply Planning															
Demand Planning		..													
Data Gathering															

Strategic or Tactical?



EXPERT INPUT

CATHY BUDD

**Director, Demand Driven Business Operations
The Dow Chemical Company**

Annual Sales: ~ \$54 billion

Main Products: Chemicals

Cathy serves as a senior consultant to operating businesses within Dow



Executive S&OP and Risk

The 21st Century Supply Chain:

- Extended
- Brittle
- Longer lead times
- More variable lead times

Supply Chain disruptions:

- Potential
- Actual

Types of Supply Chain Disruptions

- **Supply crash**
- **Demand spike**
- **Demand crash**
- **Supply spike**

Executive S&OP and Risk

The 21st Century Supply Chain:

- Extended
- Brittle
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Supply Chain disruptions:

- Potential
- Actual

Executive S&OP can help with:

- Anticipating potential disruptions
- Recovering from actual disruptions

Executive S&OP's Role in Anticipation

- **Initially set hedges:**
 - **safety stock**
 - **safety time**
 - **location**
 - **consignment**
 - **finished products**
 - **components/raw materials**
- **Review and revise hedges during the Executive S&OP cycle**
- **Permanent agenda item – Pre-Meeting and Exec Meeting**

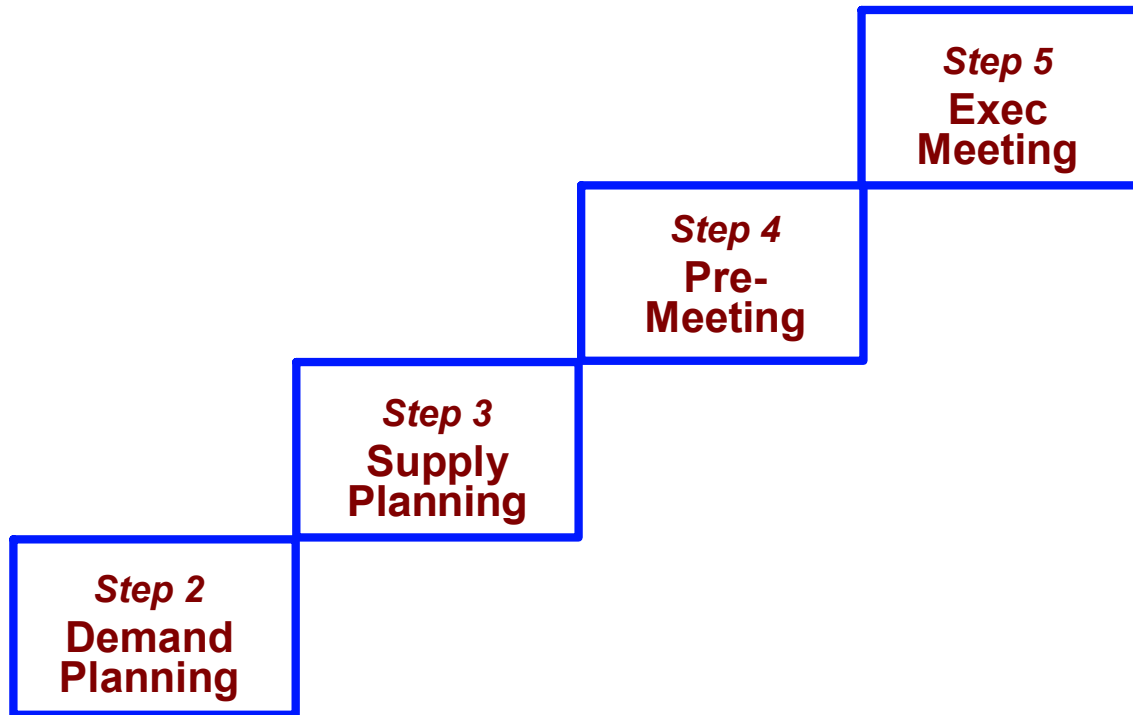
Executive S&OP's Role in Recovery

- **Minor Disruption: probably no role**
- **Moderate to Major Disruption:**

The “Mini-S&OP Cycle”

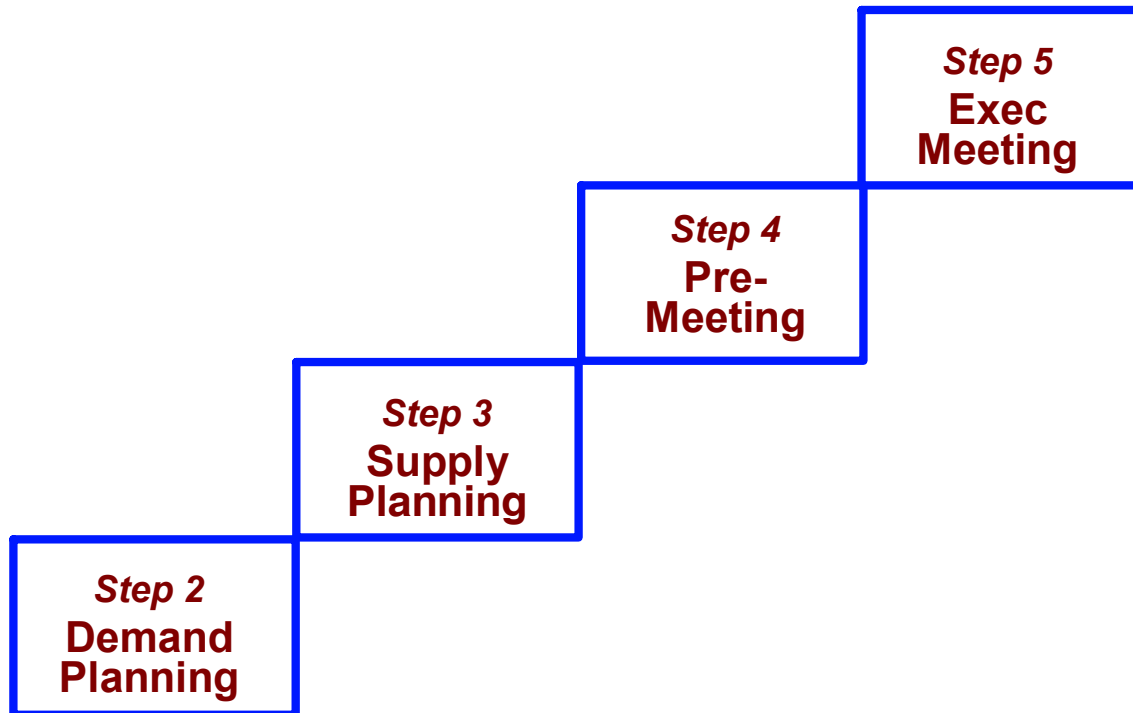
The Mini-S&OP Cycle

- Focus on affected parts of the business
- Solve the problem at the earliest step possible:



The Mini-S&OP Cycle

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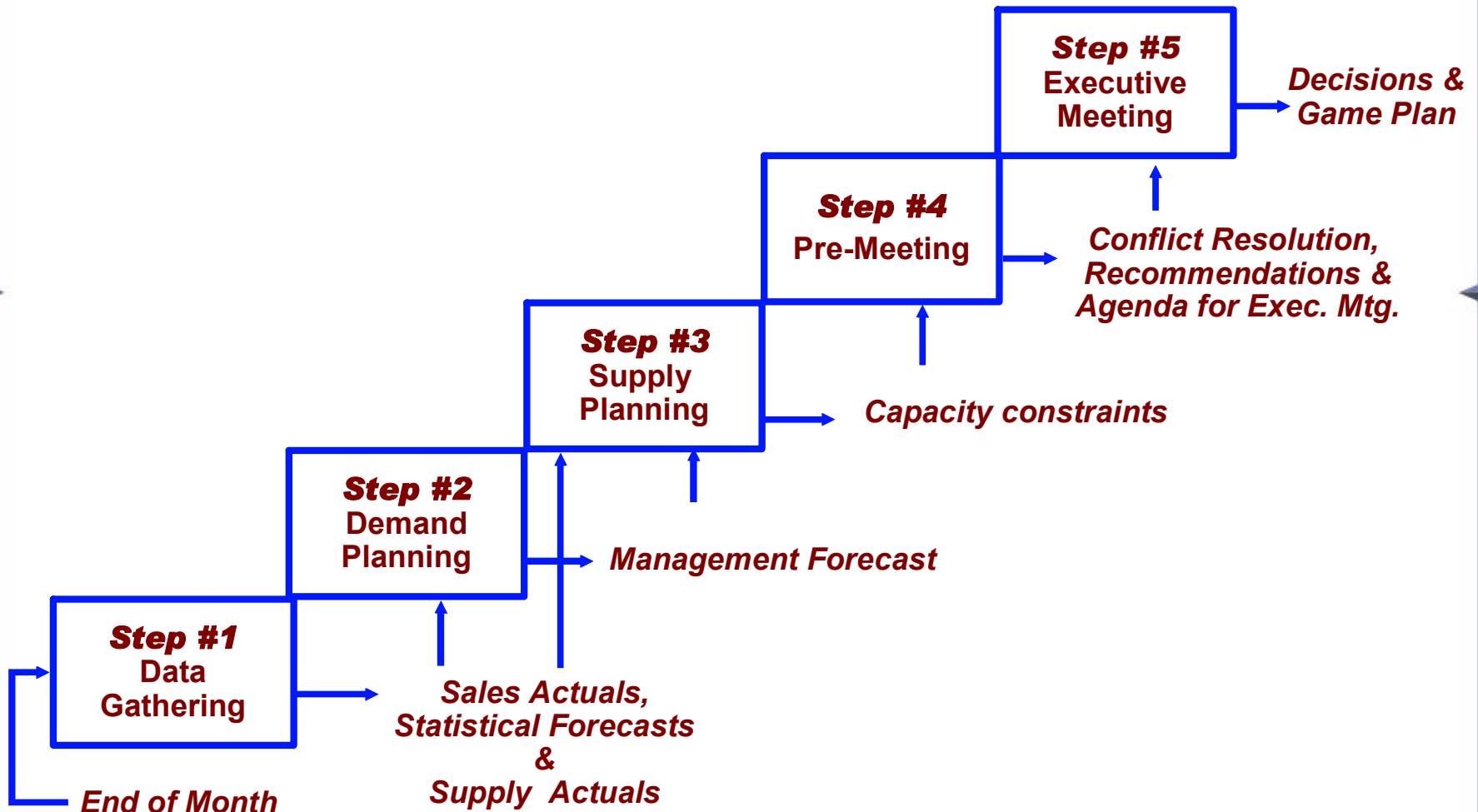


- Use the same people, report formats, and processes as in the regular monthly cycle

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Executive S&OP Process



The Global Challenge:

To coordinate an organization doing business in many different countries, on many different continents . . .

. . . without inhibiting accountability, and stifling local initiatives and ways of doing business.

Global Executive S&OP

How One Company Does It:

- Entities: areas (regions) where demand and supply align

Regions & Entities: Example

Regions

North America

South America

Europe

Middle East/Africa

Asia/Pacific

Plants

Mexico, US

Belgium

China, Australia

Entities

Regions & Entities: Example

<u>Regions</u>	<u>Plants</u>	<u>Entities</u>
North America	Mexico, US	Y
South America		N
Europe	Belgium	Y
Middle East/Africa		N
Asia/Pacific	China, Australia	Y

Global Executive S&OP

How One Company Does It:

- Entities: areas (regions) where demand and supply align
- Executive S&OP occurs within each Entity – up to and including the Exec S&OP meeting

Global Executive S&OP

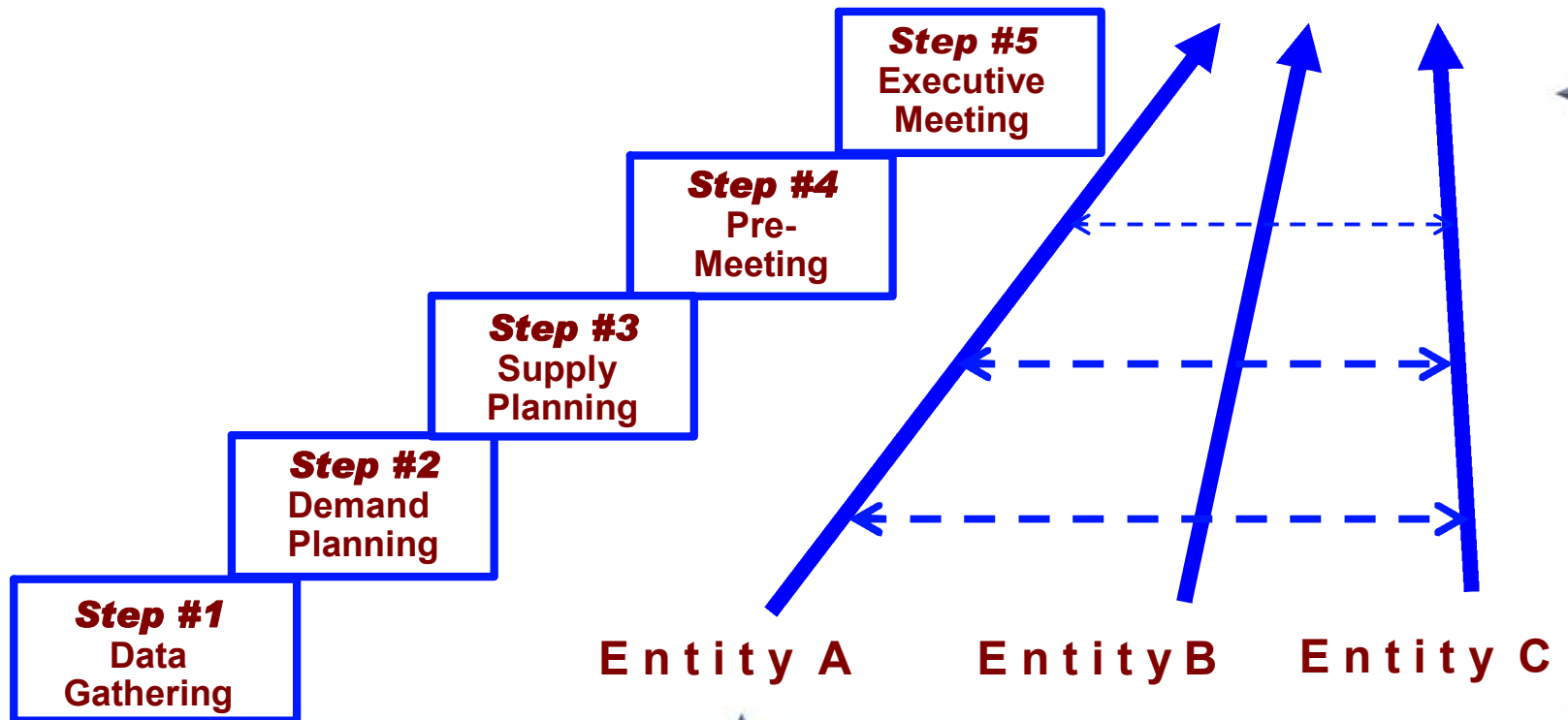


Global Executive S&OP

How One Company Does It:

- Entities: areas (regions) where demand and supply align
- Executive S&OP occurs within each Entity – up to and including the Exec S&OP meeting
- Inter-entity Coordination – primarily within Demand Planning and Supply Planning, also at Pre-Meeting

Global Executive S&OP

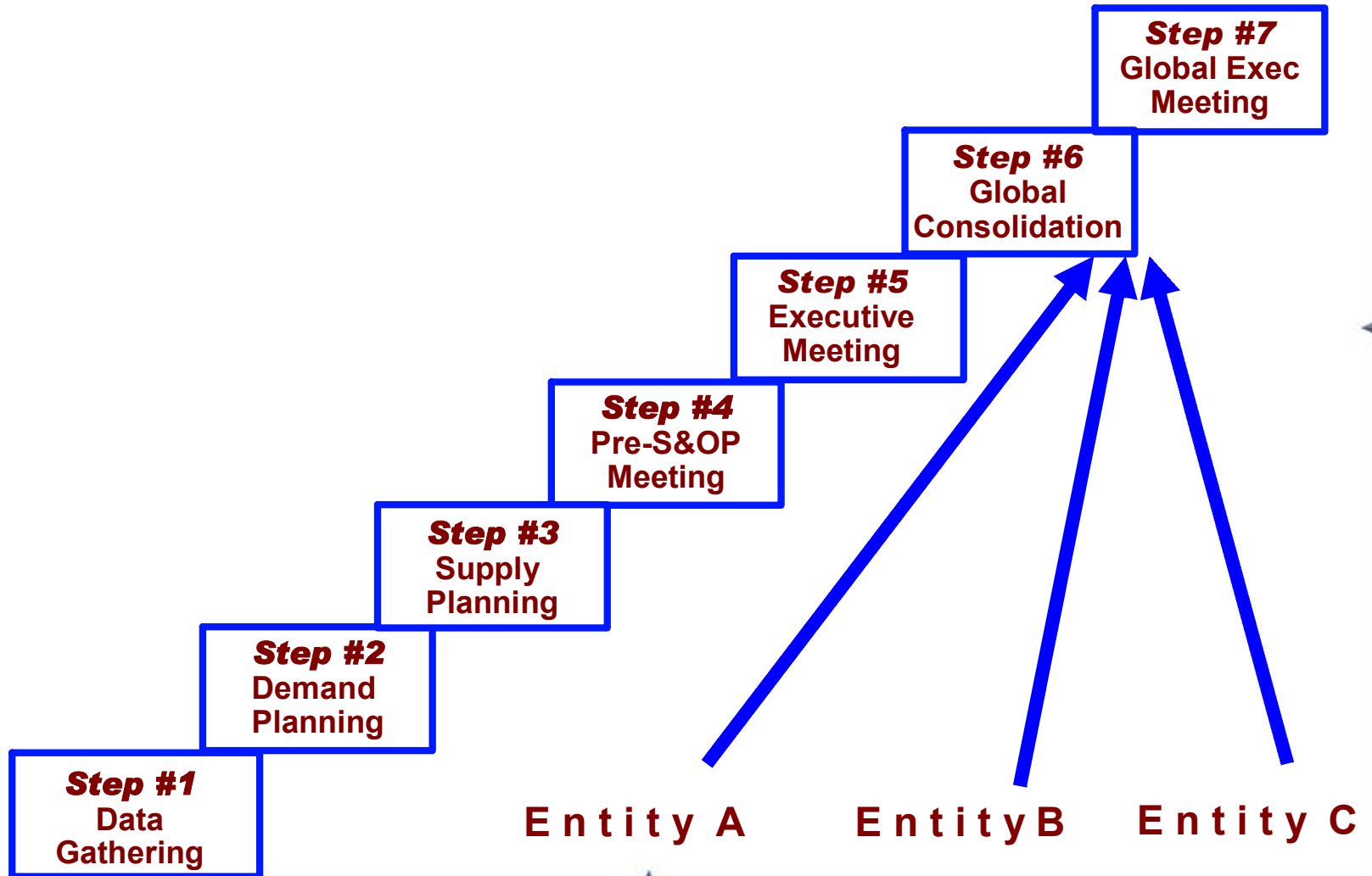


Global Executive S&OP

How One Company Does It:

- Entities: areas (regions) where demand and supply align
- Executive S&OP occurs within each Entity – up to and including the Exec S&OP meeting
- Inter-entity Coordination – primarily within the Demand Planning and Supply Planning phases
- Global Consolidation – at world HQ, following Exec meetings in the Entities
- Global Exec Meeting – primarily, but not exclusively, financial

Global Executive S&OP



Global Executive S&OP

Question:

If you sell products around the world
but have supply in only one region,
do you need Global S&OP?



Global Executive S&OP

Question:

If you sell products around the world
but have supply in only one region,
do you need Global S&OP?



The Global Challenge

“We are a series of organizations doing business locally, with intense global coordination.” – Percy Barnevik
Former CEO
ABB

Executive S&OP should support these dual objectives.

EXPERT INPUT

CONOR McDONALD

**Supply Chain Manager
Microsoft Corporation**

Annual Sales: ~ \$50 billion

**Main Products: Software, Games,
and Computer Accessories**

Conor is based in Ireland



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Implementing Executive S&OP

Bad News

- Change
- 8-12 Months

Good News

- Few People
(several dozen)
- Early Results
(2-3 Months)
- Low Cost
(< \$100K)

The ABC's of Implementation

C – Computer

B – Data

A – People

C Item - Software for S&OP

Traditional: Spreadsheet Software

- Advantages:
 - Ultra low cost
 - Quick to come up
 - No learning curve
- Disadvantages:
 - Simulation > Slow
 - Unit and Dollar Alignment > Weak
 - Volume and Mix Alignment > Weak
 - “Blood, Sweat, and Excel”
 - Dead End

Software for S&OP – Cont'd

New: S&OP Specific Software

- Disadvantages:
 - Can Slow Down the Implementation
 - Cost
 - Learning Curve
- Advantages:
 - Simulation > Rapid, Real Time
 - Unit and Dollar Alignment > Strong, Constant
 - Volume and Mix Alignment > Strong, Constant
 - Fast Preparation
 - Supports Growth and Greater Complexity

So, What To Do About Software?

Options:

1. Use spreadsheet software
2. Use S&OP specific software
3. Hybrid: start with spreadsheet
switch to specific later

When in doubt, pick 3.

B Item: DATA ISSUES

- Families and Sub-families
- Resources

Product Families and Supply Resources

Product Families

- Customers
- Markets
- Application
- How the Product is Used

What works best for Marketing and Sales

Supply Resources

- Plants
- Departments
- Suppliers
- How the Product is Made

What works best for Operations, Supply Chain, Logistics, etc.

B Item: DATA ISSUES

- Families and Sub-families
- Resources
- Units of Measure
- Load Profiles/Bills of Resources
- Average Selling Prices and Gross Margins
- Actuals: Sales, Bookings, Production, Inventories, Backlog
- Demand/Supply Strategies:
 - Where “Meet the Customer” – (MTS, MTO, FTO)
 - Target Fill Rates
 - Target Finished Goods Inv: Days On Hand
 - Target Customer Order Backlog: Days On Hand
 - Target Module Inv: Days On Hand

A Item: People

The problem is not understanding Executive S&OP. It's simple

The hard part is

Organizational Behavior Change

. . . changing how we collectively manage the business.

Issues: Discipline/Self-discipline

Accountability

Conflict Management

Implementing Executive S&OP Is Different

The president must provide:

- Support
- Funding
- Commitment
- Leadership

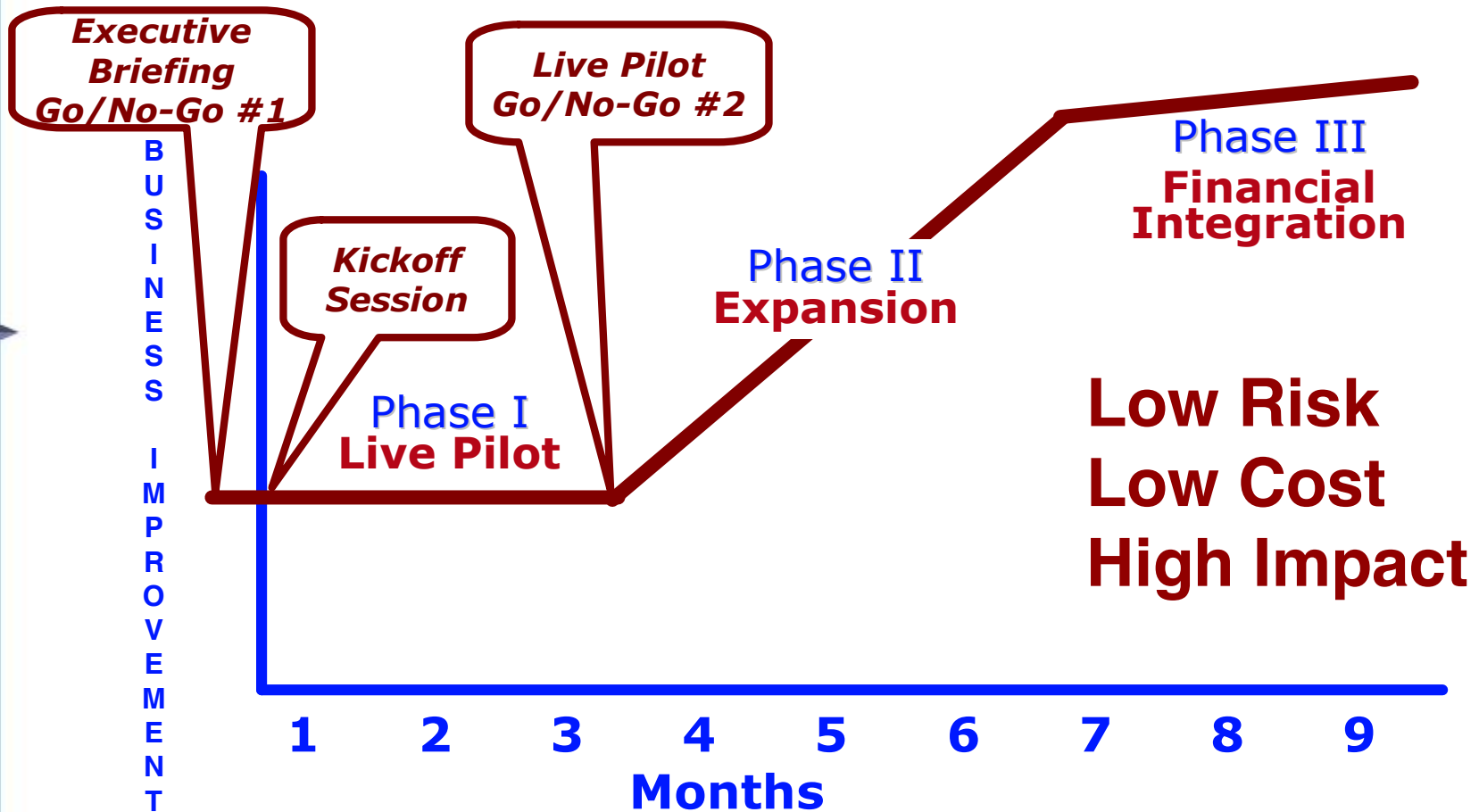
So what's new?

Active, Hands-on Participation by the President and Staff Each and Every Month

Implementation Principles

- People are the A item
- Implementation begins and ends with the Leader of the business
- “Hold the high ground” – involve top management at the very outset of the implementation, and throughout – and you will probably succeed
- “Build it and they will come” – design it first, get the mechanics working, and then try to sell it to top management – carries a lower probability for success

Implementation Path



Identifying a Broken Process

Executive S&OP is “broken” if:

- Top management is not involved
- Teamwork has not visibly improved
- Meeting attendance is poor
- Focus is on mix and the short term
- Conflict is not resolved
- Finger-pointing is not “off limits”
- Decisions are not made
- and so on

Fixing a Broken Executive S&OP Process

- Convene a group, including a member of the executive staff
- Conduct a self-audit
- Answer the question: do the President and staff participate each month?
- If yes, fix problems shown in self-audit
- If no, begin a re-implementation

Re-implementing Executive S&OP

- Will top management agree to an Executive Briefing?
- If yes, proceed on the Implementation Path
- If no, will they agree to an “Outsider Audit”?
- If they agree:
 - present audit results to top management
 - include a “mini Executive Briefing”
 - get agreement to do one product family correctly
- If no, don’t give up
- Keep it alive, improve it, and keep the executive staff member involved.

The Future of S&OP: Growth Factors

- Success breeds success
- Lean Manufacturing and S&OP
- Globalization
- New users outside traditional manufacturing
- S&OP specific software
- Greater financial integration, power and utility
- A growing presence in the executive suite

EXPERT INPUT

MARK COVAS

**Global Innovation Manager
The Procter & Gamble Company**

Annual Sales: ~ \$70 billion

Main Products: Health Care, Household Products

Mark came to P&G via the Gillette acquisition



Thanks for Being Here!

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To talk: 513-281-0500

Good Luck and Godspeed